HEMPRASATH SANKAR
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SUMMARY
Backend-focused CSE undergraduate (2028) seeking SDE Intern / Backend Engineering roles, with a 1909 LeetCode rating, three internships (including NLC India Ltd), and two first-author IEEE papers. Experienced in building REST APIs, relational/NoSQL schemas, Redis caching, BullMQ job queues, JWT/RBAC auth, and CI/CD workflows in Node.js, Python, and PostgreSQL.

TECHNICAL SKILLS
• Languages & Core CS: Java, Python, JavaScript (ES6+), TypeScript, SQL; DSA, DBMS, OS, Computer Networks
• Backend & Distributed Systems: Node.js, Express.js, FastAPI, Django, RESTful APIs, Redis (Bloom Filters, Caching, Lua Rate Limiting), JWT/RBAC, Firebase Cloud Messaging (FCM)
• Databases & Storage: PostgreSQL, MongoDB, Redis, Supabase, Relational Schema Normalization, NoSQL Document Modeling
• Frontend & Visualization: React.js, React Native (Expo), Tailwind CSS, Framer Motion, Streamlit
• ML Engineering, MLOps & Tools: Scikit-learn, Prophet, LSTM, Gradient Boosting, K-Means, MLflow, Airflow, Evidently AI, Prometheus, Grafana, Docker, Kubernetes, Git/GitHub, GitHub Actions, Postman

EXPERIENCE
Data Science / ML Intern – Zidio Technologies | github.com/hemprasath100/DSML [Mar 2026 – May 2026]
• Built an automated demand-forecasting and customer-segmentation pipeline processing 50,000+ transaction records (RFM features, Prophet, LSTM, K-Means, Gradient Boosting) and exposed batch inference results through a Streamlit analytics service.
• Containerized training and evaluation workflows with Docker/Kubernetes, Airflow, and MLflow, adding Prometheus, Grafana, and Evidently AI drift alerts to cut manual model re-evaluation time by 40% across scheduled runs.

Web Development Intern – Amdox Technologies | github.com/hemprasath100/jobPortal [Nov 2025 – Jan 2026]
• Developed a full-stack recruitment portal (React.js, Node.js, Express.js, MongoDB) with JWT authentication and role-based access control (RBAC), consolidating job postings, candidate profiles, and application tracking across 12+ recruiter workflows.
• Built RESTful APIs for candidate record management and an automated certificate generation and verification module, eliminating manual sign-off steps and reducing certificate issuance turnaround from 3 days to under 10 seconds.

Web Development Intern – NLC India Ltd (IHMS) | github.com/hemprasath100/NLCIL [Jun 2026 (2 weeks, on-site)]
• Built a PHP/MySQL cartridge-request management application for an internal department, designing the normalized relational schema and backend modules for the multi-stage request-to-approval workflow.

SELECTED PROJECTS
ShieldPay – Low-Latency UPI QR & Phishing Link Fraud Firewall | github.com/hemprasath100/shieldpay
• Built a real-time payment-link and QR-payload verification API using Node.js (TypeScript), FastAPI, Redis, and PostgreSQL to flag tampered UPI requests and SMS phishing links in under 150ms p95 latency (k6 test at 300 concurrent users).
• Implemented a two-tier lookup pipeline using Redis Bloom Filters for under 5ms O(1) screening of malicious VPAs/domains and Redis Lua sliding-window rate limiters, filtering 85% of repeat requests in synthetic replay tests before hitting workers.
• Operationalized published IEEE research by integrating a privacy-aware PII masking middleware (redacting phone numbers, bank IDs, and VPA handles) and BullMQ asynchronous worker queues with Jest/PyTest unit tests in GitHub Actions CI/CD.

CivicLens – AI-Powered Civic Issue Reporting & Triage Platform | github.com/LEE777-code/civiclenss
• Developed backend service and database layer: designed normalized Supabase/PostgreSQL schemas across complaints, citizens, municipal departments, and field workers, exposing secure REST APIs across 4 permissioned role-based workflows.
• Integrated Google Gemini Vision API server-side to classify uploaded issue photos by category and severity, achieving 82% triage accuracy and reducing manual routing time by 60% in testing on 120 sample civic complaints.
• Engineered deterministic SLA and status-tracking module (Submitted to Assigned to In Progress to Resolved) integrated with Firebase Cloud Messaging (FCM) to trigger real-time push notifications on state changes.

LaunchPilot AI – AI Growth-Strategy Prototype for App Launches | github.com/HackOnVibeCom/2026-07-high-5-hackers
• Built interactive competitor-analysis, go-to-market roadmap, and campaign-tracking dashboard UI using React, TypeScript, Tailwind CSS, and Framer Motion, implementing modular component architecture and typed state over prototype datasets.

PUBLICATIONS
• Hemprasath Sankar et al., 'Privacy-Aware Prompt Masking Interface for Secure Large Language Model Interactions,' IEEE Conference Proceedings, 2026. ieeexplore.ieee.org/document/11505205
• Hemprasath Sankar et al., 'Uncertainty-Calibrated Trust Modelling for LLM-Generated Misinformation Detection,' IEEE Conference Proceedings, 2026. ieeexplore.ieee.org/document/11505140

EDUCATION
Chennai Institute of Technology, Chennai, India [Aug 2024 – May 2028 (Expected)]
B.E. in Computer Science and Engineering – CGPA: 8.32/10

ACHIEVEMENTS
• LeetCode Knight Badge: Achieved a 1909 rating (Top 4% globally) with 600+ problems solved across data structures, graph algorithms, and dynamic programming.
• 2nd Prize – HackOVibe Hackathon: Awarded in the World Impact & Business Success Track for building LaunchPilot AI.
• 3x National/Collegiate Hackathon Finalist: Advanced to final evaluation round across three inter-collegiate hackathons.

CERTIFICATIONS
• NPTEL: IoT and Data Analytics using Python | IBM: Data Science | Cisco: Networking Essentials | ServiceNow: Training